

ASSIGNMENT NO. 1

1. PROBLEM STATEMENT:

Write a program in C to check whether a given year is a Leap Year

OBJECTIVE:

1. To understand the use of conditional statements in C programming.
2. To implement logical conditions for solving real-world problems.
3. To develop problem-solving skills using decision control structures.

THEORY:

1.Theory for conditional statements with syntax (if , if-else, if-else ladder) are used in C programming

2.Theory for a Leap Year - year that contains 366 days instead of the usual 365 days.

Rules:

- A year is a leap year if it is divisible by 400, OR
- A year is a leap year if it is divisible by 4 but not divisible by 100.

PLATFORM: Linux

CODE:

```
#include <stdio.h>

int main() {
    int year;
    printf("Enter a year: ");
    scanf("%d", &year);

    if ((year % 4 == 0 && year % 100 != 0) || (year % 400 == 0)) {
        printf("%d is a leap year.\n", year);
    } else {
        printf("%d is not a leap year.\n", year);
    }

    return 0;
}
```

OUTPUT:

```
Enter a year: 2000
2000 is a leap year.
```

CONCLUSION: Thus, the program successfully checks whether a given year is a Leap Year using conditional statements in C.